

Deep Dive: 17. OOP II: Inheritance

Concept Summary

Inheritance allows a new 'Child' class to copy all attributes and methods of an existing 'Parent' class, saving code.

Coding Structure

```
class Child(Parent):  
    # new code
```

Example Code

```
class Animal:  
    def eat(self): print('Eating')  
  
class Cat(Animal):  
    def meow(self): print('Meow')  
  
c = Cat()  
c.eat()  
c.meow()
```

Code Explanation

- Line 1-2: Creates a Parent class with a generic 'eat' method.
- Line 4-5: Creates a Child class 'Cat' that inherits from 'Animal'. It gets the 'eat' method for free!
- Line 8: Proves inheritance works: the Cat can eat().
- Line 9: The Cat can also use its own unique meow() method.

Practice Problems

1. Create a 'Vehicle' parent class with a 'start_engine' method. Create a 'Motorcycle' child class that inherits it.
2. Override the 'start_engine' method inside the child class to print a different sound (Polymorphism).